ebola — Extracted Transmission Models

Overview

A total of 513 transmission models were extracted from 232 articles that were flagged for containing extractable transmission models. The predominant model architecture was Compartmental (71.0%), with Human to human (direct contact) (69.2%) representing the most commonly modelled transmission route.

Summary

Metric	Value
Models extracted	513
Articles considered	232
Deterministic models	253 (49.3%)
Stochastic models	220 (42.9%)
Models with available code	208 (40.5%)

Figure 1: Model Overview

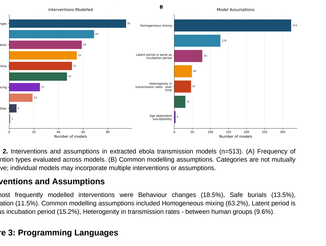

Figure 1. Overview of 513 transmission models extracted for ebola. (A) Distribution of model architectures. (B) Classification by stochastic or deterministic formulation. (C) Primary transmission routes incorporated. (D) Proportion of models with publicly available source code.

Model Formulation

Deterministic formulations were employed in 49.3% of models, while stochastic approaches accounted for 42.9%. Empirical data were incorporated in 48.3% of the extracted models, with 40.0% classified as purely theoretical.

Figure 2: Interventions and Assumptions

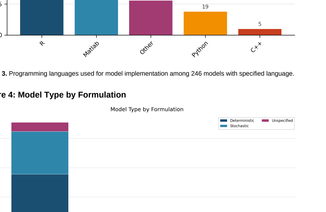

Figure 2. Interventions and assumptions in extracted ebola transmission models (n=513). (A) Frequency of intervention types evaluated across models. (B) Common modelling assumptions. Categories are not mutually exclusive; individual models may incorporate multiple interventions or assumptions.

Interventions and Assumptions

The most frequently modelled interventions were Behaviour changes (18.5%), Safe burials (13.5%), Vaccination (11.5%). Common modelling assumptions included Homogeneous mixing (63.2%), Latent period is same as incubation period (15.2%), Heterogeneity in transmission rates - between human groups (9.6%).

Figure 3: Programming Languages

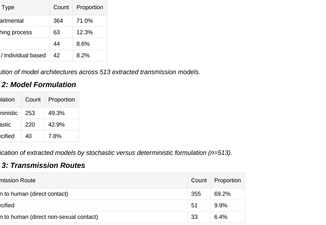

Figure 3. Programming languages used for model implementation among 246 models with specified language.

Figure 4: Model Type by Formulation

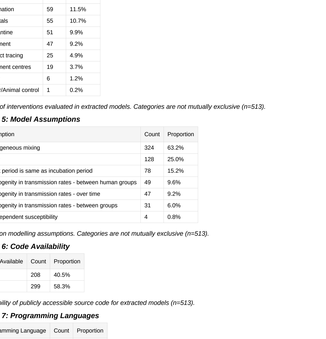

Figure 4. Distribution of deterministic and stochastic formulations across model types (n=513).

Reproducibility

Source code was publicly available for 40.5% of the extracted models.

Tables

Table 1: Model Types

Model Type	Count	Proportion
Compartmental	364	71.0%
Branching process	63	12.3%
Other	44	8.6%
Agent / Individual based	42	8.2%

Distribution of model architectures across 513 extracted transmission models.

Table 2: Model Formulation

Formulation	Count	Proportion
Deterministic	253	49.3%
Stochastic	220	42.9%
Unspecified	40	7.8%

Classification of extracted models by stochastic versus deterministic formulation (n=513).

Table 3: Transmission Routes

Transmission Route	Count	Proportion
Human to human (direct contact)	355	69.2%
Unspecified	51	9.9%
Human to human (direct non-sexual contact)	33	6.4%
Airborne or close contact	19	3.7%
Human to human (direct contact);Vector/Animal to human	17	3.3%
Human to human (direct contact);Sexual	7	1.4%
Human to human (direct non-sexual contact);Sexual	6	1.2%
Human to human (direct contact);Unspecified	6	1.2%
Vector/Animal to human	5	1.0%
Vector/Animal to human;Human to human (direct contact)	4	0.8%
Human to human (direct contact);Airborne or close contact	2	0.4%
Sexual	1	0.2%
Human to human (direct non-sexual contact);Airborne or close contact	1	0.2%
Airborne or close contact;Human to human (direct non-sexual contact);Sexual	1	0.2%
Human to human (direct contact);Vector/Animal to human;Sexual	1	0.2%
Human to human (direct non-sexual contact);Vector/Animal to human	1	0.2%
Human to human (direct contact);Human to human (direct non-sexual contact)	1	0.2%
Human to human (direct contact);Sexual;Vector/Animal to human	1	0.2%
Airborne or close contact;Human to human (direct contact)	1	0.2%

Primary transmission routes incorporated in extracted models (n=513).

Table 4: Interventions Modelled

Intervention Type	Count	Proportion
Behaviour changes	95	18.5%
Safe burials	69	13.5%
Vaccination	59	11.5%
Hospitals	55	10.7%
Quarantine	51	9.9%
Treatment	47	9.2%
Contact tracing	25	4.9%
Treatment centres	19	3.7%
Other	6	1.2%
Vector/Animal control	1	0.2%

Types of interventions evaluated in extracted models. Categories are not mutually exclusive (n=513).

Table 5: Model Assumptions

Assumption	Count	Proportion
Homogeneous mixing	324	63.2%
Other	128	25.0%
Latent period is same as incubation period	78	15.2%
Heterogenity in transmission rates - between human groups	49	9.6%
Heterogenity in transmission rates - over time	47	9.2%
Heterogenity in transmission rates - between groups	31	6.0%
Age dependent susceptibility	4	0.8%

Common modelling assumptions. Categories are not mutually exclusive (n=513).

Table 6: Code Availability

Code Available	Count	Proportion
Yes	208	40.5%
No	299	58.3%

Availability of publicly accessible source code for extracted models (n=513).

Table 7: Programming Languages

Programming Language	Count	Proportion
Unspecified	267	52.0%
R	124	24.2%
Matlab	70	13.6%
Other	28	5.5%
Python	19	3.7%
C++	5	1.0%

Programming languages used for model implementation (n=513).

Table 8: Empirical Data Usage

Empirical Data Used	Count	Proportion
Yes	248	48.3%
No/Unspecified	265	51.7%

Model Inventory (Sample)

Table 9 presents a sample of 10 extracted model records. The complete dataset of 513 extracted transmission models for ebola is publicly available.

Table 9: Sample of Extracted Models

Article ID	Model Type	Compartmental Structure	Formulation	Transmission Route
PMID_16999875	Compartmental	Other compartmental	Stochastic	Human to human (direct contact)
PMID_28193880	Agent / Individual based	Not compartmental	Stochastic	Human to human (direct contact)
PMID_28193880	Agent / Individual based	Not compartmental	Stochastic	Human to human (direct contact)
PMID_25642364	Compartmental	SEIR	Deterministic	Human to human (direct contact)
PMID_25833863	Compartmental	SEIR	Deterministic	Human to human (direct contact)
PMID_25833863	Compartmental	SEIR	Stochastic	Human to human (direct contact)
PMID_26886846	Compartmental	Other compartmental	Stochastic	Vector/Animal to human
PMID_25736239	Compartmental	SEIR	Deterministic	Human to human (direct non-sexual contact)
PMID_25736239	Compartmental	Other compartmental	Deterministic	Human to human (direct non-sexual contact)
PMID_25736239	Compartmental	Other compartmental	Deterministic	Human to human (direct non-sexual contact)

Sample of 10 records from 513 extracted ebola transmission models.

Data Availability

The complete dataset of extracted transmission models for ebola (n=513) has been made publicly available to support future modelling efforts and outbreak preparedness.